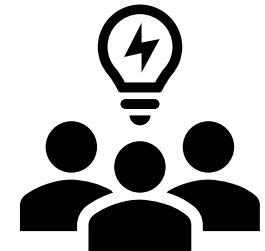
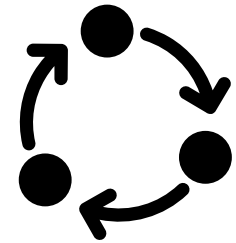


ML - Driven Customer Segmentation and Retention Agent

(Amazon Prime)



Data Mining, Project Proposal

Instructor: Shun Shi

Meet Our Team

Group 03



Roshni S Rana
(400683499)



Problem Statement



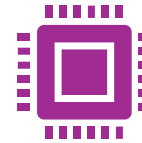
Customer retention is a major challenge for digital platforms such as **streaming services, e-commerce, and subscription businesses.**



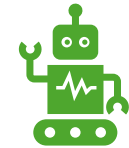
Many companies **lose customers due to the inability to identify behavioral patterns early.**



Large volumes of customer data are collected, but **actionable insights are often not extracted effectively.**



Without early detection of **at-risk customers**, businesses face **revenue loss and reduced customer lifetime value.**



Machine learning can help **analyze behavioral data, segment customers, and predict churn risk** to support better retention strategies.

Why We Chose This Problem?

We selected this problem because:

- From a **data science perspective**, this project is interesting because it combines:
- **Customer segmentation**
- **Predictive analytics**
- **Business decision support**
- Customer churn directly impacts company revenue and Profits
- This allows us to apply **machine learning techniques Practically with Real Life Problems and End to End ML models.**



AIM

- In today's digital economy, **subscription and platform businesses focus heavily on churn, customer lifetime value, and user behavior.**
- Platforms such as **Netflix, Spotify, Amazon Prime, Slack, Amazon, Shopify, Uber, and Lyft** rely on understanding customer activity to retain users.
- Businesses that can **analyze user behavior, segment customers intelligently, and predict churn** gain a strong competitive advantage.
- Therefore, this project aims to **build an AI-driven Customer Segmentation and Retention Agent that analyzes user activity data to identify customer segments and predict users who are likely to leave the platform on early stage to support revenue of the business.**

The YouTube logo, featuring the word "You" in black and "Tube" in white inside a red rounded rectangle.The Amazon Prime logo, with the word "prime" in blue lowercase letters and the Amazon smile arrow below it.The Netflix logo, a red vertical bar.The Spotify logo, featuring a green circular icon with three curved lines and the word "Spotify" in green.The Google One logo, with "Google" in its multi-colored font and "One" in grey.

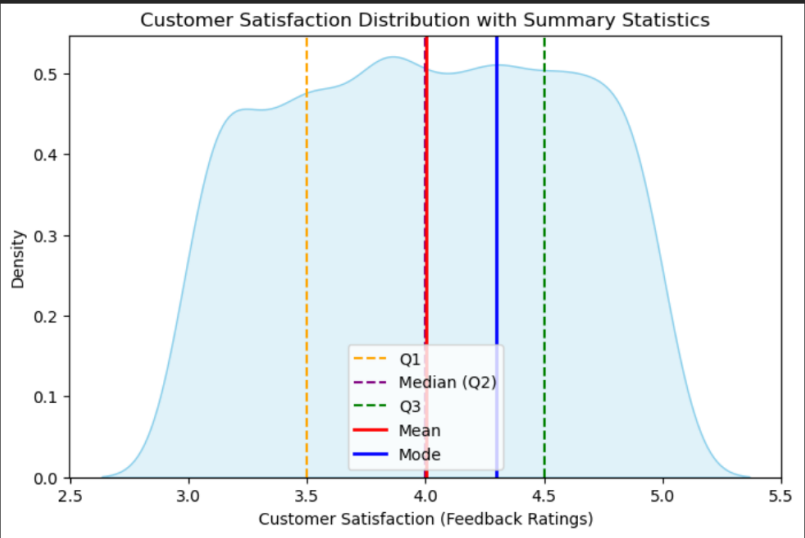
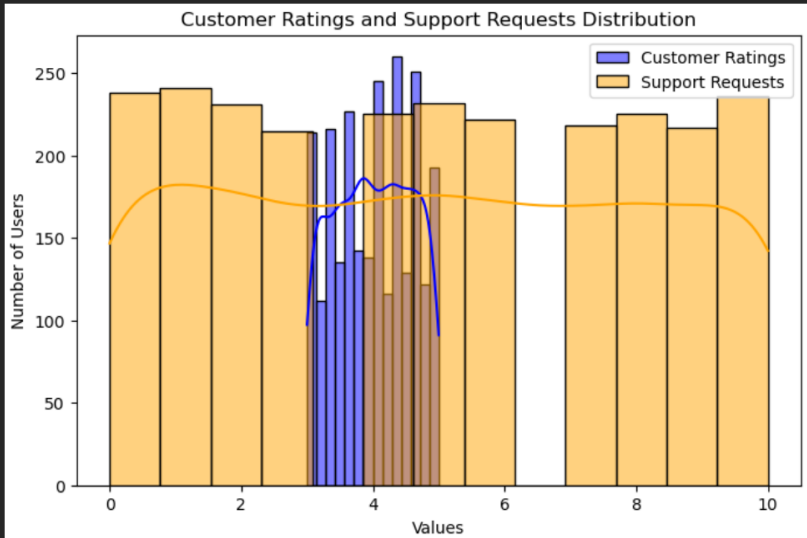
Data Set

Amazon Prime User Dataset

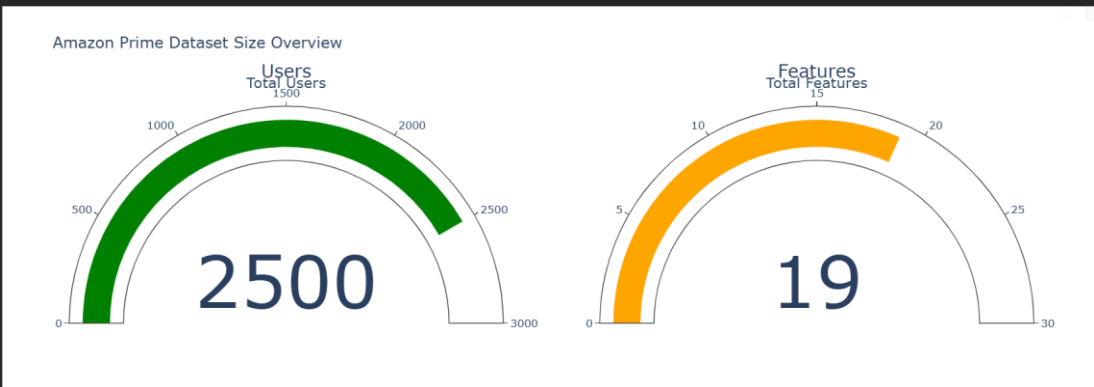
We will use the **Amazon Prime User Dataset** which contains information about user behavior From The Kaggle platform.



Data Size, Shape and Summary Stats



	Statistic	Customer Satisfaction	Service Usage Behavior
0	Mean	4.004760	4.951600
1	Median (Q2)	4.000000	5.000000
2	Mode	4.300000	1.000000
3	Q1 (25%)	3.500000	2.000000
4	Q3 (75%)	4.500000	8.000000
5	Minimum	3.000000	0.000000
6	Maximum	5.000000	10.000000
7	Standard Deviation	0.580769	3.191572

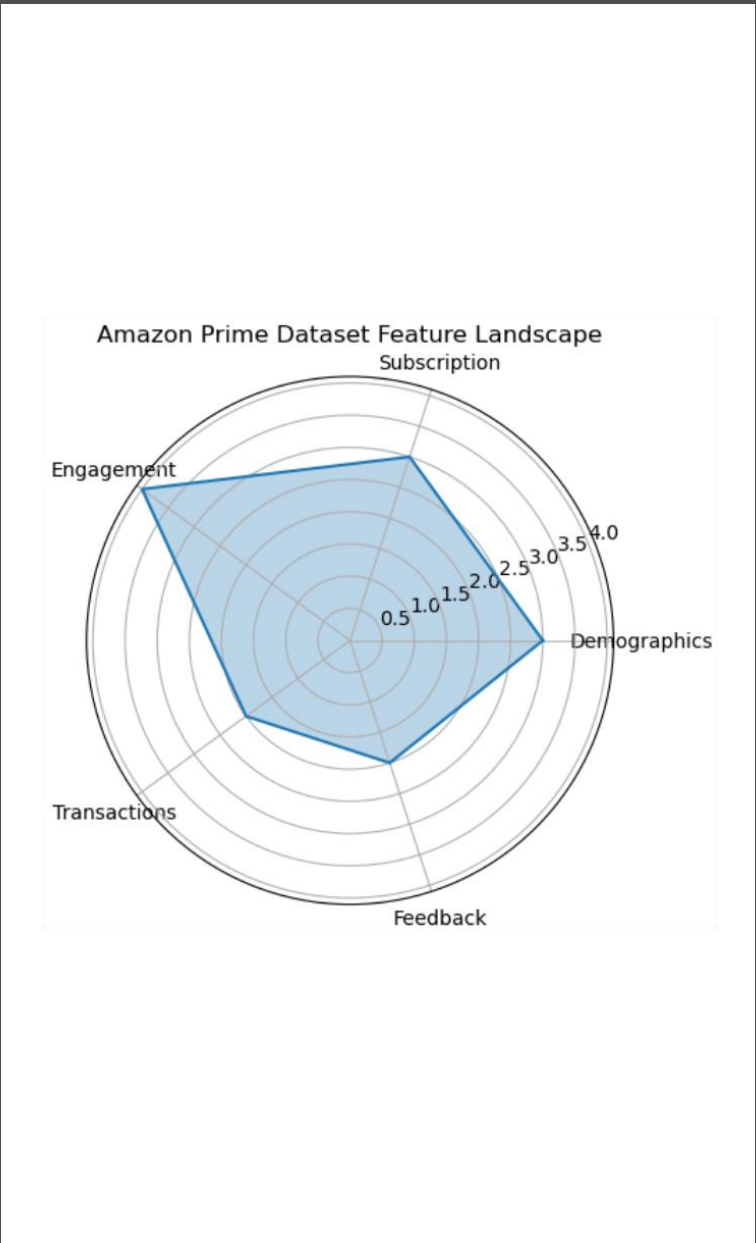


Why this Dataset?



	Total Rows	No Null Values	Null Values
user_id	2500	2500	0
name	2500	2500	0
email_address	2500	2500	0
username	2500	2500	0
date_of_birth	2500	2500	0
gender	2500	2500	0
location	2500	2500	0
membership_start_date	2500	2500	0
membership_end_date	2500	2500	0
subscription_plan	2500	2500	0
payment_information	2500	2500	0
renewal_status	2500	2500	0
usage_frequency	2500	2500	0
purchase_history	2500	2500	0
favorite_genres	2500	2500	0
devices_used	2500	2500	0
engagement_metrics	2500	2500	0
feedback_ratings	2500	2500	0
customer_support_interactions	2500	2500	0

- Dataset (19 features)
- Demographics
 - gender
 - location
 - date_of_birth
 - Subscription
 - membership_start_date
 - membership_end_date
 - subscription_plan
 - renewal_status
 - Engagement
 - usage_frequency
 - favorite_genres
 - devices_used
 - engagement_metrics
 - Transactions
 - payment_information
 - purchase_history
 - Feedback
 - feedback_ratings
 - customer_support_interactions



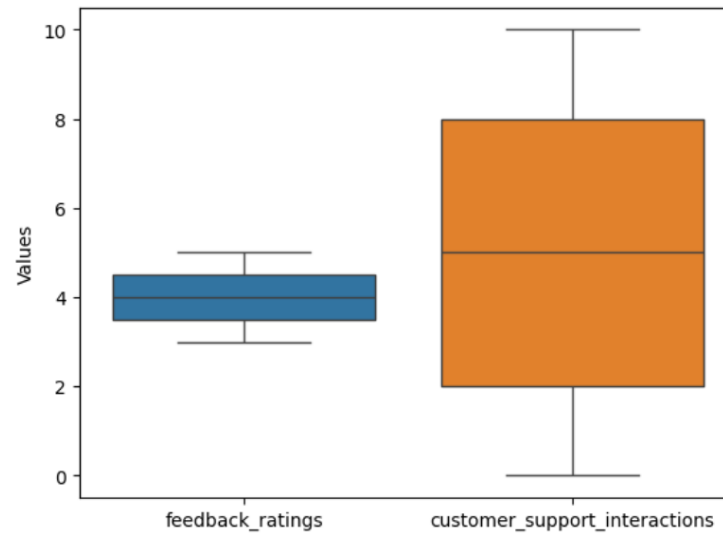
Amazon Prime Dataset

Rows(2500)

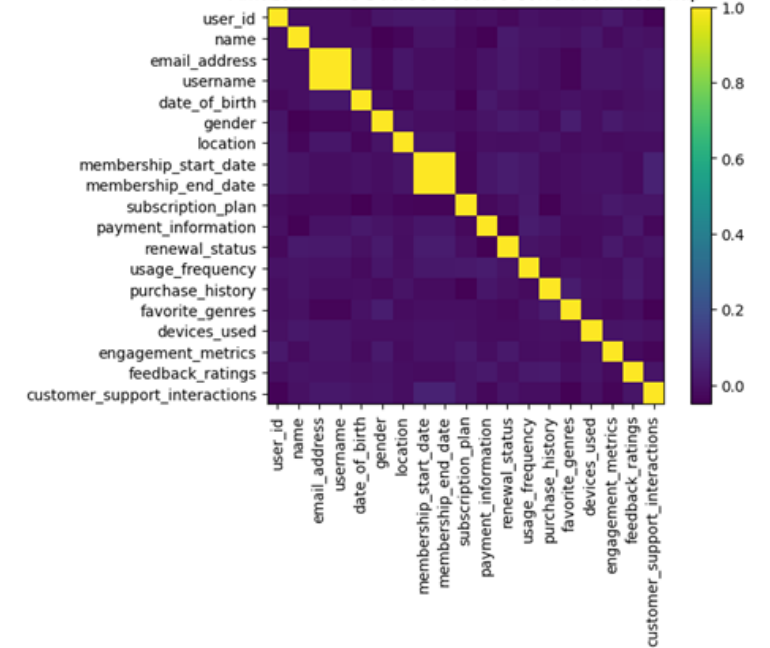
Columns(19)

```
user_id
name
email_address
username
date_of_birth
gender
location
membership_start_date
membership_end_date
subscription_plan
payment_information
renewal_status
usage_frequency
purchase_history
favorite_genres
devices_used
engagement_metrics
feedback_ratings
customer_support_interactions
```

Outlier Detection



Amazon Prime Dataset Feature Correlation Heatmap



Customer Churn Prediction Objective and Modeling Plan

Objective

- Our goal is to build a machine learning model that **identifies customers who is about to leave platform.**
- By detecting these users early, companies can **take retention actions and reduce revenue loss.**

Classification Objective

We will classify customers into two groups:

Class 0 – Loyal / Active Customers

Customers who continue using the platform regularly.

Class 1 – At-Risk / Potential Churn Customers

Customers who show behavioral signals indicating they may leave the platform.

Evaluation Metrics

To evaluate model performance, we will focus on:

Accuracy – Measures the overall correctness of predictions.

Precision – Ensures the model correctly identifies true churn customers.

Recall – Captures as many churn cases as possible.

Key Insight:

Recall is especially important because missing churn customers means losing potential revenue.

Algorithms We are Planning to Use



K-Means Clustering
Used for customer segmentation to group users with similar behavior patterns.



Random Forest
A powerful classification model that captures complex relationships in customer data.



Logistic Regression
Used as a baseline model for churn prediction and model comparison.



These algorithms allow us to compare different approaches and select the most effective model for predicting customer churn.

Expected Output

- The final system will provide:
- Customer segments based on user behavior and usage patterns
- Churn risk prediction for each customer
- Insights to help businesses design targeted retention strategies
- This can help companies improve **customer satisfaction, retention rates, and long-term revenue growth.**

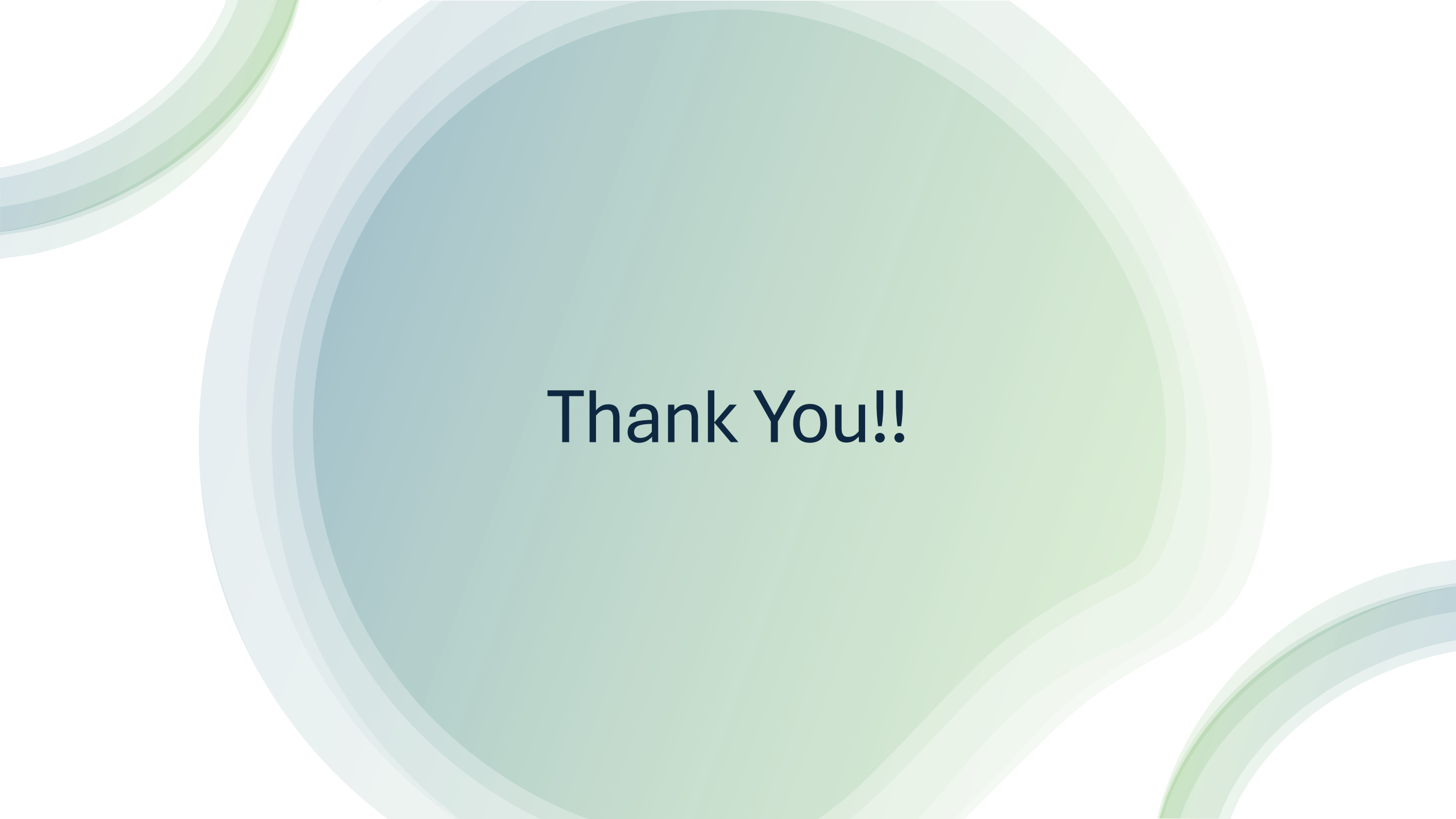




Future Scope

This project can be extended in several ways to enhance its real-world impact:

- **Real-Time Churn Prediction**
Develop real-time systems that continuously monitor user behavior and detect customers who are at risk of leaving the platform.
- **Integration with Recommendation Systems**
Combine churn prediction with recommendation engines to suggest personalized content, products, or services that increase engagement.
- **Personalized Marketing Strategies**
Use customer behavior insights to design targeted promotions and retention campaigns.
- **Cloud-Based Deployment**
Deploy the model on cloud platforms to support scalable and automated decision-making systems.
- **AWS AI Retention Agent**
Implement an AI-powered retention agent using AWS services that automatically identifies at-risk customers and triggers retention actions such as personalized offers or engagement messages.
- **Advanced Modeling with Deep Learning**
Apply deep learning models to capture complex behavioral patterns and improve churn prediction accuracy.
- **CRM Integration**
In the future, the system can integrate with CRM platforms to automatically alert businesses when a customer shows signs of potential churn.



Thank You!!